ECE 3210 hw09

1. Given the differential equation

$$(D^2 + 7D + 12)y(t) = (3D + 9)f(t),$$

where $y(0^-) = 1$, $y'(0^-) = 2$, and $f(t) = e^{-3t}u(t)$, find the following responses using the Laplace transform.

(a) Find the zero-input response $y_{\text{zi}}(t)$.

Solution: For the zero-input response set $f(t) = 0$ and take the Laplace transform, using $y(0^-) = 1$ and $y'(0^-) = 2$ in the derivative terms. Then solve for $Y(s)$ and factor the denominator for partial fractions.

$$s^2Y(s) - s - 2 + 7\big(sY(s) - 1\big) + 12Y(s) = 0$$
$$\implies Y(s) = \frac{s+9}{s^2+7s+12} = \frac{a}{s+3} + \frac{b}{s+4}$$

The residues follow from the cover-up method:

$$a = (s+3)Y(s)\Big|_{s=-3} = \frac{6}{1} = 6$$
$$b = (s+4)Y(s)\Big|_{s=-4} = \frac{5}{-1} = -5$$

Inverting each term gives

$$y_{\text{zi}}(t) = \big(6e^{-3t} - 5e^{-4t}\big)u(t)$$

(b) Find the zero-state response $y_{\text{zs}}(t)$.

Solution: With zero initial conditions, the transfer function comes straight from the differential equation, and the input transform is that of $e^{-3t}u(t)$:

$$H(s) = \frac{3s+9}{s^2+7s+12}, \qquad F(s) = \frac{1}{s+3}$$

The zero-state output is the product $H(s)F(s)$. Factoring the numerator as $3s + 9 = 3(s + 3)$ and expanding in partial fractions gives

$$\begin{aligned} Y(s) &= H(s)F(s) \\ &= \frac{3(s+3)}{s^2+7s+12} \cdot \frac{1}{s+3} \\ &= \frac{3}{s+3} - \frac{3}{s+4} \end{aligned}$$

Inverting each term,

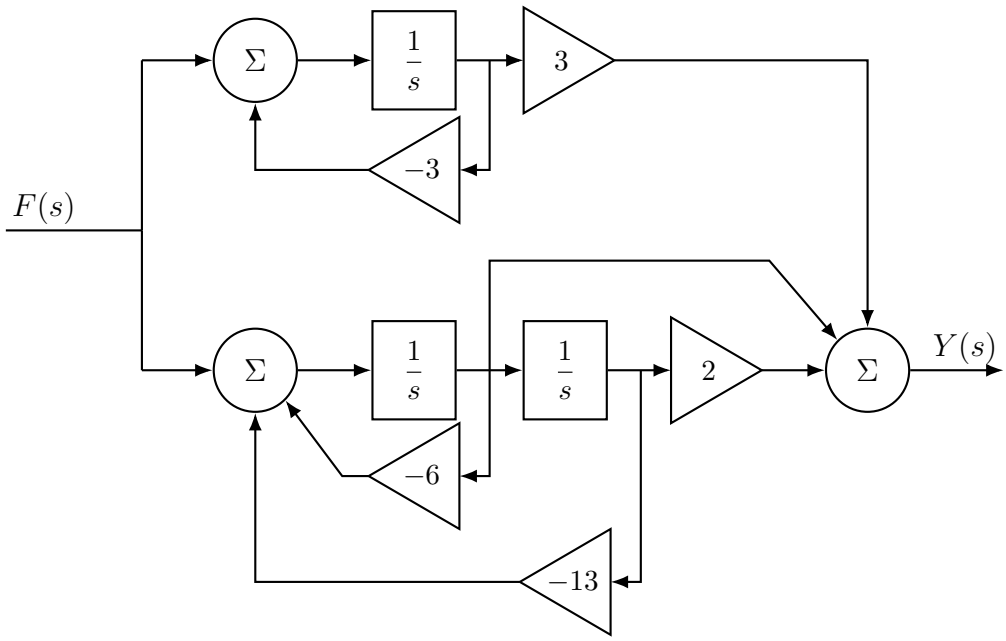
$$y_{\text{zs}}(t) = \left(3e^{-3t} - 3e^{-4t}\right)u(t)$$

(c) Find the total response $y_{\text{tot}}(t)$.

Solution: The total response is the sum of the zero-input and zero-state responses:

$$\begin{aligned} y_{\text{tot}}(t) &= \left(6e^{-3t} - 5e^{-4t}\right)u(t) + \left(3e^{-3t} - 3e^{-4t}\right)u(t) \\ &= \left(9e^{-3t} - 8e^{-4t}\right)u(t) \end{aligned}$$

2. Given the parallel system realization below, find the system's impulse response $h(t)$.



Solution: Reading the two parallel branches off the realization, the output is the sum of each branch applied to $F(s)$. Dividing out $F(s)$ gives the transfer function.

$$Y(s) = \frac{3F(s)}{s+3} + \frac{s+2}{s^2+6s+13}F(s)$$
$$H(s) = \frac{3}{s+3} + \frac{s+2}{s^2+6s+13}$$

The quadratic does not factor over the reals, so complete the square, $s^2+6s+13 = (s+3)^2+2^2$, and rewrite the numerator as $s+2 = (s+3)-1$ to match the cosine and sine transform pairs:

$$H(s) = \frac{3}{s+3} + \frac{s+3}{(s+3)^2+2^2} + \frac{2}{(s+3)^2+2^2}\cdot\left(-\frac{1}{2}\right)$$

Inverting term by term,

$$h(t) = e^{-3t}\left(3+\cos(2t)-\tfrac{1}{2}\sin(2t)\right)u(t)$$

3. Given the differential equation

$$(D^2 + 6D + 25)y(t) = (D + 1)f(t),$$

where $y(0^-) = 1$, $y'(0^-) = 2$, and $f(t) = u(t)$, find the following responses using the Laplace transform.

(a) Find the zero-input response $y_{\text{zi}}(t)$.

Solution: Set $f(t) = 0$ and transform, using $y(0^-) = 1$ and $y'(0^-) = 2$. The denominator has complex roots, so complete the square to $(s+3)^2 + 4^2$.

$$s^2Y(s) - s - 2 + 6\big(sY(s) - 1\big) + 25Y(s) = 0$$

$$Y(s) = \frac{s+8}{s^2+6s+25} = \frac{s+8}{(s+3)^2+4^2}$$

Split the numerator as $(s+3)+5$ so that the terms match the damped cosine and damped sine transform pairs:

$$Y(s) = \frac{s+3}{(s+3)^2+4^2} + \frac{4}{(s+3)^2+4^2} \cdot \frac{5}{4}$$

Inverting,

$$y_{\text{zi}}(t) = e^{-3t}\left(\cos(4t) + \tfrac{5}{4}\sin(4t)\right)u(t)$$

(b) Find the zero-state response $y_{\text{zs}}(t)$.

Solution: With zero initial conditions, the transfer function is read from the differential equation, and the input is a unit step:

$$H(s) = \frac{s+1}{s^2+6s+25}, \qquad F(s) = \frac{1}{s}$$

Then $Y(s) = H(s)F(s)$ has one real pole at $s = 0$ and a complex-conjugate pair at $s = -3 \pm j4$, so the partial-fraction form has a conjugate pair of residues b and b^*:

$$Y(s) = \frac{a}{s} + \frac{b}{s+3-j4} + \frac{b^*}{s+3+j4}$$

Evaluating the residues gives

$$Y(s) = \frac{0.04}{s} + \frac{-0.02 - j0.11}{s+3-j4} + \frac{-0.02 + j0.11}{s+3+j4}$$

The conjugate pair combines into a damped cosine and sine (the real part of b sets the cosine coefficient and the imaginary part sets the sine coefficient):

$$y_{\text{zs}}(t) = \Big(0.04 + e^{-3t}\big(-0.04\cos(4t) + 0.22\sin(4t)\big)\Big)u(t)$$

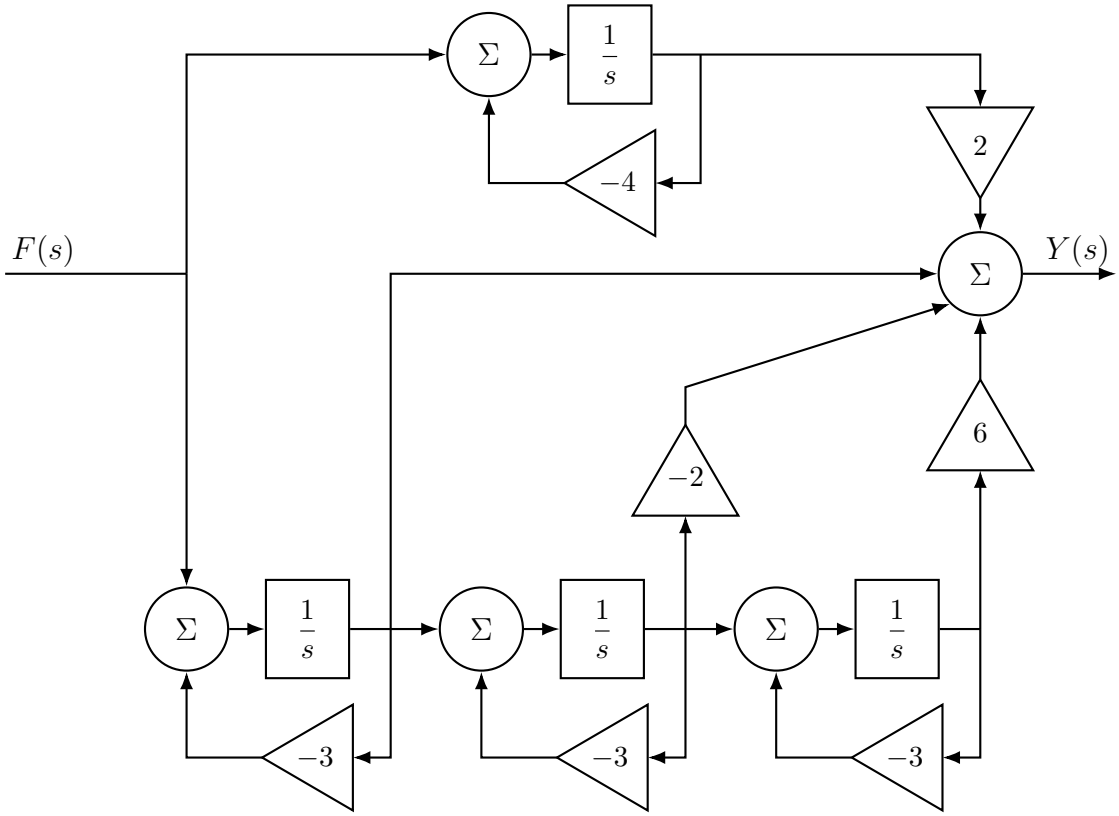
(that is, $\frac{1}{25}$, $-\frac{1}{25}$, and $\frac{11}{50}$).

(c) Find the total response $y_{\text{tot}}(t)$.

Solution: The total response is the sum of the zero-input and zero-state responses:

$$\begin{aligned} y(t) &= y_{\text{zi}}(t) + y_{\text{zs}}(t) \\ &= \Big(0.04 + e^{-3t}\big(0.96\cos(4t) + 1.47\sin(4t)\big)\Big)u(t) \end{aligned}$$

4. Given the system realization below, find the system's impulse response $h(t)$.



Solution: Reading each path of the realization, the output is the sum of the path responses applied to $F(s)$, with poles at $s = -4$ and a repeated $s = -3$. Dividing out $F(s)$ gives the transfer function:

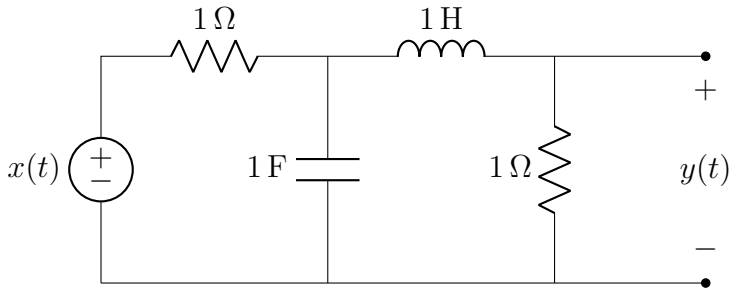
$$Y(s) = \frac{2}{s+4}F(s) + \frac{1}{s+3}F(s) - \frac{2F(s)}{(s+3)^2} + \frac{6F(s)}{(s+3)^3}$$

$$H(s) = \frac{2}{s+4} + \frac{1}{s+3} - \frac{2}{(s+3)^2} + \frac{6}{(s+3)^3}$$

Each term is already in a standard form ($\frac{1}{s+a} \to e^{-at}$, $\frac{1}{(s+a)^2} \to te^{-at}$, $\frac{1}{(s+a)^3} \to \frac{t^2}{2}e^{-at}$), so the inverse transform can be written directly:

$$h(t) = \left(2e^{-4t} + e^{-3t} - 2te^{-3t} + 3t^2e^{-3t}\right)u(t)$$

5. Consider the circuit below.



(a) Derive the transfer function $H(s)$ for this circuit.

Solution: Label the node between the $1\,\Omega$ resistor and the inductor $V(s)$. In the s-domain the capacitor has impedance $1/s$ and the inductor has impedance s. KCL at that node gives

$$\frac{V(s)-X(s)}{1}+\frac{V(s)}{1/s}+\frac{V(s)-Y(s)}{s}=0$$

The output $Y(s)$ is the voltage across the $1\,\Omega$ resistor, so a voltage divider relates it to $V(s)$:

$$Y(s)=\frac{V(s)\cdot 1}{1+s} \implies V(s)=(1+s)Y(s)$$

Expand KCL and collect the $V(s)$ terms:

$$\implies V(s)-X(s)+sV(s)+\frac{V(s)}{s}-\frac{Y(s)}{s}=0$$
$$\implies V(s)\left(1+s+\frac{1}{s}\right)-X(s)-\frac{Y(s)}{s}=0$$

Substitute $V(s)=(1+s)Y(s)$ to eliminate $V(s)$ and expand the product:

$$\implies Y(s)(1+s)\left(1+s+\frac{1}{s}\right)-X(s)-\frac{Y(s)}{s}=0$$
$$\implies Y(s)\left(1+s+\frac{1}{s}+s+s^2+1\right)-X(s)-\frac{Y(s)}{s}=0$$

The $\pm\frac{1}{s}$ terms cancel, leaving $Y(s)(s^2+2s+2)=X(s)$, so

$$\implies \frac{Y(s)}{X(s)}=H(s)=\frac{1}{s^2+2s+2}$$

(b) Given the input $x(t) = e^{-t}u(t)$, find the zero-state response $y(t)$.

Solution: The input transform is that of a decaying exponential:

$$X(s) = \frac{1}{s+1}$$

Multiply by $H(s)$. The quadratic has roots $s = -1 \pm j$, so alongside the real pole at $s = -1$ there is a conjugate pair of poles, giving a conjugate pair of residues b and b^*:

$$\begin{aligned} Y(s) = H(s)X(s) &= \frac{1}{s^2+2s+2} \cdot \frac{1}{s+1} \\ &= \frac{a}{s+1} + \frac{b}{s+1-j} + \frac{b^*}{s+1+j} \end{aligned}$$

Evaluating the residues:

$$Y(s) = \frac{1}{s+1} + \frac{-0.5}{s+1-j} + \frac{-0.5}{s+1+j}$$

The conjugate pair of residues is real here, so it combines into a damped cosine:

$$y(t) = e^{-t}\big(1 - \cos(t)\big)u(t)$$

6. Consider the differential equation

$$(D^2 + 4D + 4)y(t) = (2D^2 + 11D + 8)f(t),$$

with $y(0^-) = 4$, $y'(0^-) = -6$, and input $f(t) = u(t)$. Find the following responses using the Laplace transform.

(a) Find the zero-input response $y_{\text{zi}}(t)$.

Solution: Set $f(t) = 0$ and transform, using $y(0^-) = 4$ and $y'(0^-) = -6$. The denominator is a repeated root, $s^2 + 4s + 4 = (s+2)^2$, so the partial-fraction form needs both $\frac{1}{(s+2)^2}$ and $\frac{1}{s+2}$ terms.

$$s^2Y(s) - 4s + 6 + 4\big(sY(s) - 4\big) + 4Y(s) = 0$$
$$\implies Y(s) = \frac{4s+10}{(s+2)^2} = \frac{a}{(s+2)^2} + \frac{b}{s+2}$$

The coefficient of the highest-order pole comes from the cover-up method:

$$a = (s+2)^2 Y(s)\Big|_{s=-2} = 2$$

The cover-up method does not give b, so multiply both sides by s and take $s \to \infty$. Only the terms that do not vanish survive:

$$\implies \lim_{s\to\infty} \frac{4s^2 + 10s}{(s+2)^2} = \lim_{s\to\infty} \frac{s \cdot 2}{(s+2)^2} + \frac{b \cdot s}{s+2} \implies b = 4$$

Substituting both coefficients and inverting ($\frac{1}{(s+2)^2} \to te^{-2t}$):

$$\implies Y(s) = \frac{2}{(s+2)^2} + \frac{4}{s+2}$$
$$y_{\text{zi}}(t) = (2t+4)e^{-2t}u(t)$$

(b) Find the zero-state response $y_{\text{zs}}(t)$.

Solution: With zero initial conditions the transfer function is read from the differential equation, and the input is a unit step, $F(s) = 1/s$:

$$H(s) = \frac{2s^2 + 11s + 8}{(s+2)^2}$$

The product $H(s)F(s)$ has a simple pole at $s = 0$ and a repeated pole at $s = -2$:

$$Y(s) = \frac{2s^2 + 11s + 8}{(s+2)^2} \cdot \frac{1}{s} = \frac{a}{s} + \frac{b}{(s+2)^2} + \frac{c}{s+2}$$

The cover-up method gives a and b:

$$a = sY(s)\Big|_{s=0} = 2$$
$$b = (s+2)^2 Y(s)\Big|_{s=-2} = \frac{-6}{-2} = 3$$

As before, c comes from multiplying by s and letting $s \to \infty$:

$$\lim_{s\to\infty} s \cdot \frac{2s^2+11s+8}{(s+2)^2\, s} = \frac{s\cdot 2}{s} + \frac{s\cdot 3}{(s+2)^2} + \frac{s\cdot c}{s+2}$$
$$\implies 2 = 2 + c \implies c = 0$$

Since $c = 0$, only two terms remain, and inverting gives the response:

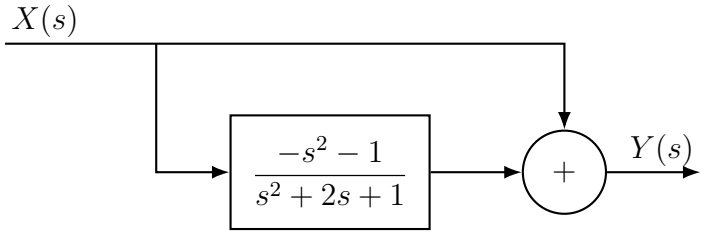
$$Y(s) = \frac{2}{s} + \frac{3}{(s+2)^2}$$
$$\implies y_{\text{zs}}(t) = \big(2 + 3te^{-2t}\big)u(t)$$

(c) Find the total response $y_{\text{tot}}(t)$.

Solution: The total response is the sum of the zero-input and zero-state responses:

$$y_{\text{tot}}(t) = \big(2 + e^{-2t}(5t+4)\big)u(t)$$

7. Consider the block diagram below. Find the impulse response $h(t)$ for this system.



Solution: The direct path and the block are summed at the output, so the output is $F(s)$ plus the block applied to $F(s)$. The block's gain is $-\frac{s^2+1}{s^2+2s+1}$, which gives the minus sign below:

$$Y(s) = F(s) - \frac{s^2+1}{s^2+2s+1}F(s)$$
$$H(s) = 1 - \frac{s^2+1}{s^2+2s+1}$$

The numerator and denominator have the same degree, so first divide to pull out the constant term: $\dfrac{s^2+1}{s^2+2s+1} = 1 + \dfrac{-2s}{s^2+2s+1}$, since

$$s^2 + 1 - (s^2 + 2s + 1) = -2s.$$

Substitute this into $H(s)$; the constants cancel, leaving a proper fraction with a repeated pole at $s = -1$:

$$\implies H(s) = 1 - \left(1 + \frac{-2s}{s^2+2s+1}\right) = \frac{2s}{s^2+2s+1} = \frac{2s}{(s+1)^2} = \frac{a}{(s+1)^2} + \frac{b}{s+1}$$

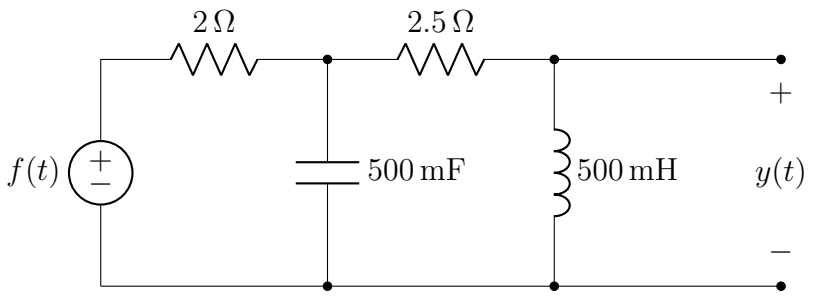
The cover-up method gives a; then b comes from multiplying by s and letting $s \to \infty$:

$$a = (s+1)^2 H(s)\Big|_{s=-1} = -2$$
$$\lim_{s\to\infty} \frac{2s^2}{(s+1)^2} = \frac{-2\cdot s}{(s+1)^2} + \frac{s\cdot b}{s+1} \implies 2 = b$$

Inverting ($\frac{1}{(s+1)^2} \to te^{-t}$) and factoring:

$$H(s) = \frac{-2}{(s+1)^2} + \frac{2}{s+1}$$
$$h(t) = \left(-2te^{-t} + 2e^{-t}\right)u(t) = 2e^{-t}(1-t)u(t)$$

8. Consider the circuit below.



(a) Derive the transfer function $H(s)$ for this circuit.

Solution: Label the node after the $2\,\Omega$ resistor $V(s)$. In the s-domain the 500 mF capacitor has impedance $2/s$, and the 500 mH inductor has impedance $0.5s$. KCL at that node:

$$\frac{V(s)-F(s)}{2}+\frac{V(s)}{2/s}+\frac{V(s)-Y(s)}{2.5}=0$$

The output is the voltage across the inductor, so a voltage divider with the $2.5\,\Omega$ resistor relates $Y(s)$ to $V(s)$. Solve it for $V(s)$:

$$Y(s)=\frac{V(s)\cdot 0.5s}{2.5+0.5s} \implies Y(s)(2.5+0.5s)=V(s)\cdot 0.5s$$
$$\implies Y(s)\,\frac{5+s}{s}=V(s)$$

Collect the $V(s)$ terms in KCL and clear the fractions by multiplying by 10:

$$\implies V(s)\left(\frac{1}{2}+\frac{s}{2}+\frac{1}{2.5}\right)-\frac{F(s)}{2}-\frac{Y(s)}{2.5}=0$$
$$\implies V(s)(5+5s+4)-5F(s)-4Y(s)=0$$

Substitute $V(s)=Y(s)(5+s)/s$ and multiply through by s to clear the remaining fraction:

$$\implies Y(s)(5+s)(5+5s+4)-5sF(s)-4sY(s)=0$$
$$\implies Y(s)\big(25+25s+20+5s+5s^2+4s\big)-5sF(s)-4sY(s)=0$$

The $+4s$ inside the bracket cancels the $-4sY(s)$ term. Solving for the ratio $Y(s)/F(s)$ and dividing by 5:

$$\implies \frac{Y(s)}{F(s)}=\frac{5s}{5s^2+30s+45}=\frac{s}{s^2+6s+9}=H(s)$$

(b) Given the input $x(t) = u(t)$, find the zero-state response $y(t)$.

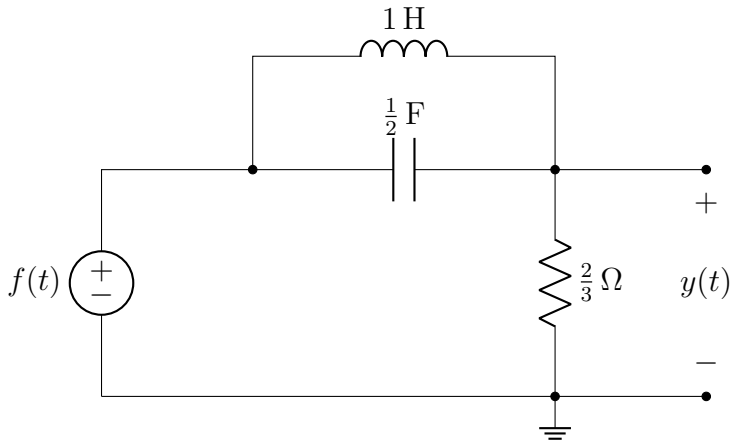
Solution: Multiply the input transform $F(s) = 1/s$ by $H(s)$; the factor of s in the numerator cancels the pole at the origin, leaving a repeated pole:

$$Y(s) = H(s)F(s) = \frac{s}{s^2 + 6s + 9} \cdot \frac{1}{s} = \frac{1}{(s+3)^2}$$

Inverting ($\frac{1}{(s+3)^2} \rightarrow te^{-3t}$):

$$y(t) = te^{-3t}u(t)$$

9. Consider the circuit below.



(a) Derive the transfer function $H(s)$ for this circuit.

Solution: The output node voltage is $Y(s)$. The capacitor has impedance $\frac{1}{(1/2)s} = 2/s$, the inductor has impedance s, and the $\frac{2}{3}\,\Omega$ resistor goes to ground. The capacitor and inductor both sit between $F(s)$ and the output node, so KCL there gives

$$\frac{Y(s) - F(s)}{2/s} + \frac{Y(s) - F(s)}{s} + \frac{Y(s)}{2/3} = 0$$

Collect the $Y(s)$ and $F(s)$ terms:

$$\Longrightarrow Y(s)\left(\frac{s}{2} + \frac{1}{s} + \frac{3}{2}\right) - F(s)\left(\frac{s}{2} + \frac{1}{s}\right) = 0$$

Multiply through by $2s$ to clear the fractions, then solve for the ratio and factor the denominator:

$$\Longrightarrow Y(s)\big(s^2 + 2 + 3s\big) - F(s)\big(s^2 + 2\big) = 0$$
$$\Longrightarrow H(s) = \frac{s^2 + 2}{s^2 + 3s + 2} = \frac{s^2 + 2}{(s+2)(s+1)}$$

(b) What type of filter is this?

Solution: Check the magnitude of $H(j\omega)$ at the two extremes of frequency:

$$\text{If } \omega \to \infty,\ H(j\omega) \to 1$$
$$\text{If } \omega \to 0,\ H(j\omega) = 1$$

Both extremes pass with unity gain, but the numerator $s^2 + 2$ vanishes at $s = \pm j\sqrt{2}$, so the response dips to zero in between:

$$\Longrightarrow \text{notch / band-reject}$$

(c) Find the impulse response $h(t)$.

Solution: The numerator and denominator have the same degree, so $H(s)$ is not a proper fraction. Divide first to pull out the constant term (this is what produces the $\delta(t)$ term in the impulse response): $\dfrac{s^2+2}{s^2+3s+2} = 1 + \dfrac{-3s}{s^2+3s+2}$, since

$$s^2 + 2 - (s^2 + 3s + 2) = -3s.$$

Write the remainder as a proper fraction, called $\tilde{H}(s)$, and expand it in partial fractions:

$$\Longrightarrow H(s) = 1 - \frac{3s}{(s+2)(s+1)}$$
$$\tilde{H}(s) = \frac{-3s}{(s+2)(s+1)} = \frac{a}{s+2} + \frac{b}{s+1}$$

The residues come from the cover-up method:

$$a = (s+2)\tilde{H}(s)\Big|_{s=-2} = \frac{6}{-1} = -6$$
$$b = (s+1)\tilde{H}(s)\Big|_{s=-1} = \frac{3}{1} = 3$$

Substitute them back and invert ($1 \to \delta(t)$):

$$\Longrightarrow H(s) = 1 - \frac{6}{s+2} + \frac{3}{s+1}$$
$$h(t) = \delta(t) + \left(-6e^{-2t} + 3e^{-t}\right)u(t)$$

(d) Given the input $f(t) = u(t)$, find the zero-state response $y(t)$.

Solution: For a unit step input, $F(s) = 1/s$ adds a pole at the origin, giving three simple poles:

$$Y(s) = H(s)F(s) = \frac{s^2+2}{(s+1)(s+2)} \cdot \frac{1}{s} = \frac{a}{s} + \frac{b}{s+1} + \frac{c}{s+2}$$

Each residue follows from the cover-up method:

$$a = sY(s)\Big|_{s=0} = 1$$
$$b = (s+1)Y(s)\Big|_{s=-1} = \frac{3}{(1)(-1)} = -3$$
$$c = (s+2)Y(s)\Big|_{s=-2} = \frac{6}{(-1)(-2)} = 3$$

Substitute the residues and invert term by term:

$$Y(s) = \frac{1}{s} - \frac{3}{s+1} + \frac{3}{s+2}$$
$$y(t) = \left(1 - 3e^{-t} + 3e^{-2t}\right)u(t)$$